

Statement of participation

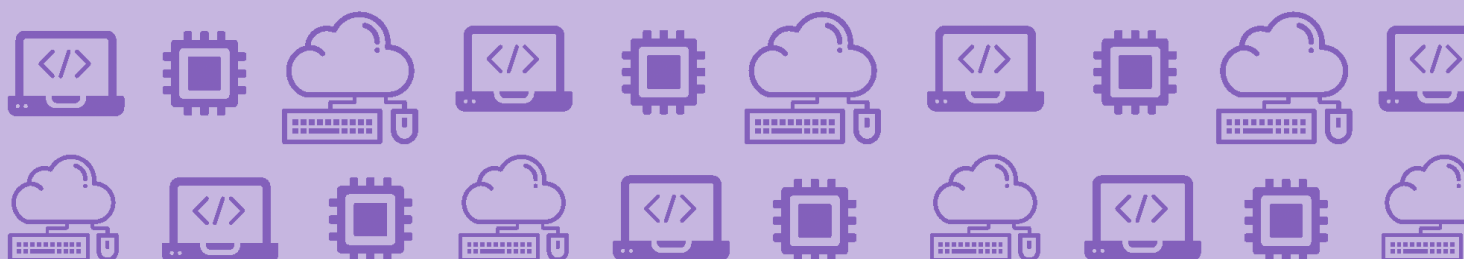
Islam Emad Ahmed Ismail

has completed the free course including any mandatory tests for:

Exploring communications technology

This free 10-hour course explores the principles underpinning communications technology.

Issue date: 13 October 2025



www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:
<https://www.open.edu/openlearn/digital-computing/exploring-communications-technology/content-section-0>

COURSE CODE: **TM355_1**

Exploring communications technology

<https://www.open.edu/openlearn/digital-computing/exploring-communications-technology/content-section-0>

Course summary

This free course, Exploring communications technology, looks at some of the principles underpinning communications technology. It looks at the properties of signals and media (such as optical fibre and radio waves), error control, data compression, and the principles of 4G mobile communications, wi-fi and DSL broadband.

Learning outcomes

By completing this course, the learner should be able to:

- demonstrate an understanding of the principles of, and reasons for, carrier-wave modulation, and its application to Quadrature Amplitude Modulation
- demonstrate an understanding of the roles of error detection and error correction, and be able to perform calculations involving check digits and coding rates
- demonstrate an understanding of the basic techniques of perceptual coding and the reasons for their use
- demonstrate an understanding of how the use of multiple OFDM subchannels enables efficient use of communications channels in mobile communications, broadband and wifi.

Completed study

The learner has completed the following:

Section 1

Signals and modulation

Section 2

Error control

Section 3

Perceptual source-coding and lossy compression

Section 4

Broadband, mobile and WiFi